Lecture Transcript by Slide

Week 10 · Chapter 8: Ventricular Rhythms · 1 h 11 min

Auto-transcribed, so expect occasional errors, especially in technical terms. Timestamps are positions in the recording.

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0:02:27 T Wave Configuration in PVCs

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0:10:45 Interpolated PVC

0:11:14 Unifocal PVCs

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0:42:43 Rules for Premature Ventricular Complex

0:44:51 Premature Ventricular Complex (ECG strip)

0:51:11 Mechanism of Ventricular Tachycardia

0:52:26 Rules for Ventricular Tachycardia

0:53:54 Ventricular Tachycardia (ECG strip)

0:58:23 Mechanism of Ventricular Fibrillation

0:59:57 Rules for Ventricular Fibrillation

1:00:25 Ventricular Fibrillation (ECG strip)

1:01:02 Mechanism of Idioventricular Rhythm

1:03:02 Rules for Idioventricular Rhythm

1:03:42 Idioventricular Rhythm (ECG strip)

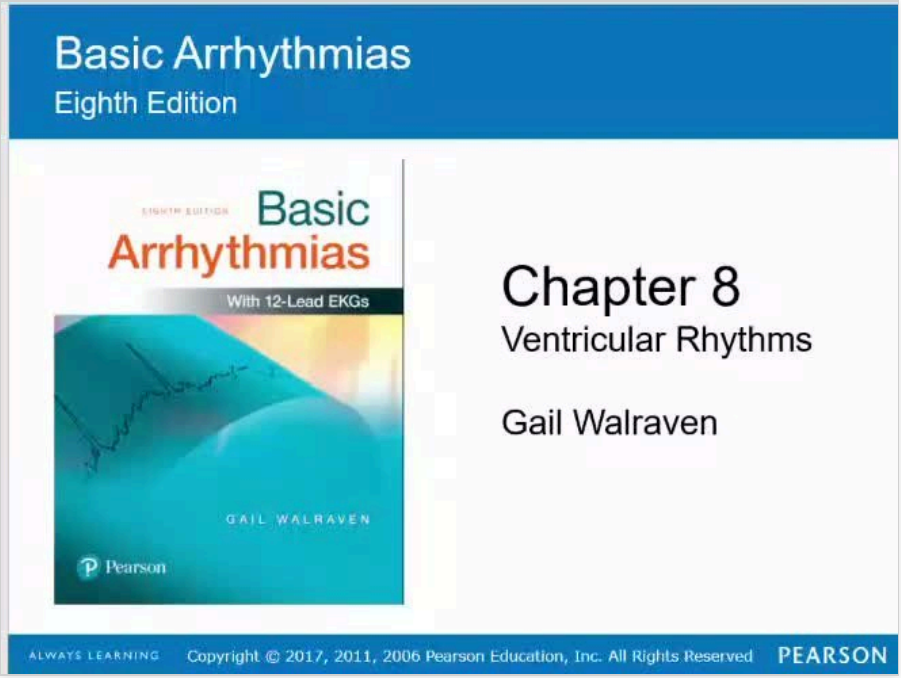
1:07:23 Mechanism of Asystole

1:08:16 Asystole (ECG strip)

1:09:19 Agonal Rhythm

1 Chapter 8: Ventricular Rhythms (title slide)

0:00:00 – 0:00:47

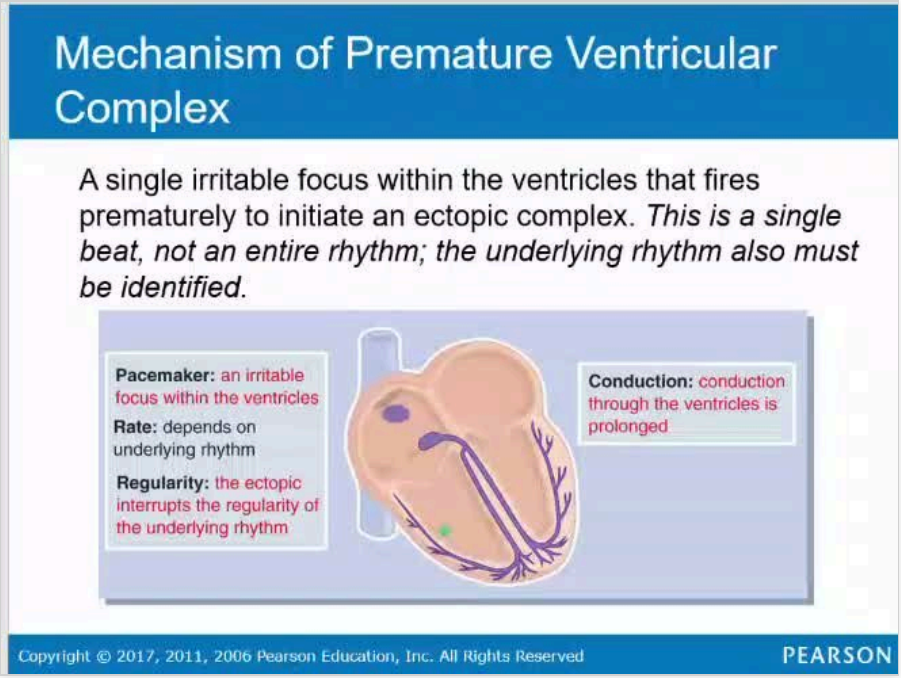


0:00:01 Okay, so this week we're going to talk about ventricular rhythms, chapter 8. I'm going to go a little bit out of order here. The book goes to Heart Blocks this week. I like to teach ventricular rhythms before I teach the Heart Blocks because the Heart Blocks are a little more challenging. Ventricular rhythms are challenging as well. So these next two weeks, ventricular rhythms this week and Heart Block rhythms next week are going to be the most two challenging subject topics that we're going to cover for the rest of the semester.

0:00:44 So the first thing we're going to talk about here is we're

2 Mechanism of Premature Ventricular Complex

0:00:47 – 0:02:27



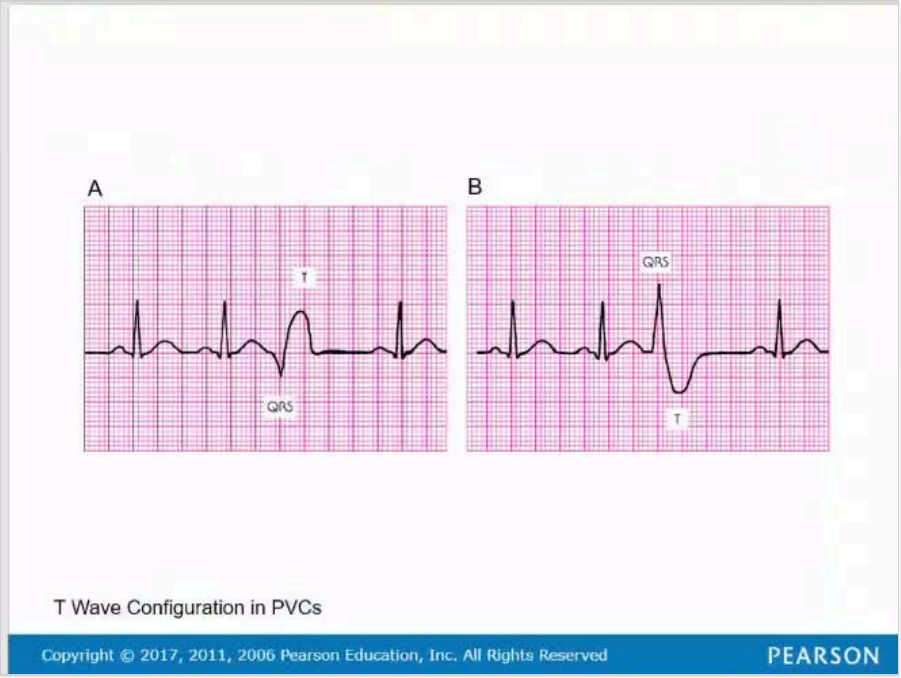
0:00:50 going to talk about premature ventricular complexes. So in previous weeks, we've talked about premature atrial complexes and premature junctional complexes. These premature ventricular complexes are the same and they are generated again by an irritable focal point, but now they are generated from an irritable focal point from within the ventricles. So as you can see, the green star down here in the right ventricle, we have a spot there that becomes irritable and it fires early or prematurely before the SA node can fire again and this is going to produce what we call a premature ventricular complex.

0:01:50 So this is a single beat, not an entire rhythm, so we're still going to have to identify the underlying rhythm like we did with PACs and PJCs.

0:02:10 So again, this is an irritable focal point that produces an ectopic impulse, fires early and causes what we call a premature ventricular complex.

3 T Wave Configuration in PVCs

0:02:27 – 0:06:46



0:02:28 So there's multiple different ways that the PVC can appear. So what we're seeing in slide A here is an underlying beat for the first two and then we see this early beat and this is a premature ventricular complex.

0:02:52 It's early because you can see how close it comes to the preceding beat, but then the other thing that happens here, and we'll talk about the rules for PVCs, but because this beat is generated from the ventricles, this beat will not be preceded by a P wave. So there's no P wave in these PVCs because the beat is coming from the bottom up. So there's no way that we're going to see a P wave preceding each of these PVCs. The other thing that we're going to discuss here is because that impulse comes from the ventricles, the impulse itself does not travel along nerve fibers. That impulse has to travel from cell to cell, myocardial cell to myocardial cell, and as a result of that, it does not spread as fast as it would if it was traveling along the nerve fibers.

0:03:55 So remember an EKG is electrical activity recorded on paper over time. So if the impulse travels fast and depolarization occurs quickly as a result, when we record that on paper from left to right, the impulse as it gets recorded on paper will be narrow. So all of the supraventricular beats, the sinus beats, the atrial beats, and the junctional beats, all of those beats travel along the nerve pathway, and so depolarization happens quickly so the QRS is narrow, 0.12 seconds or less. The PVC beats, because it's traveling from cell to cell, it travels

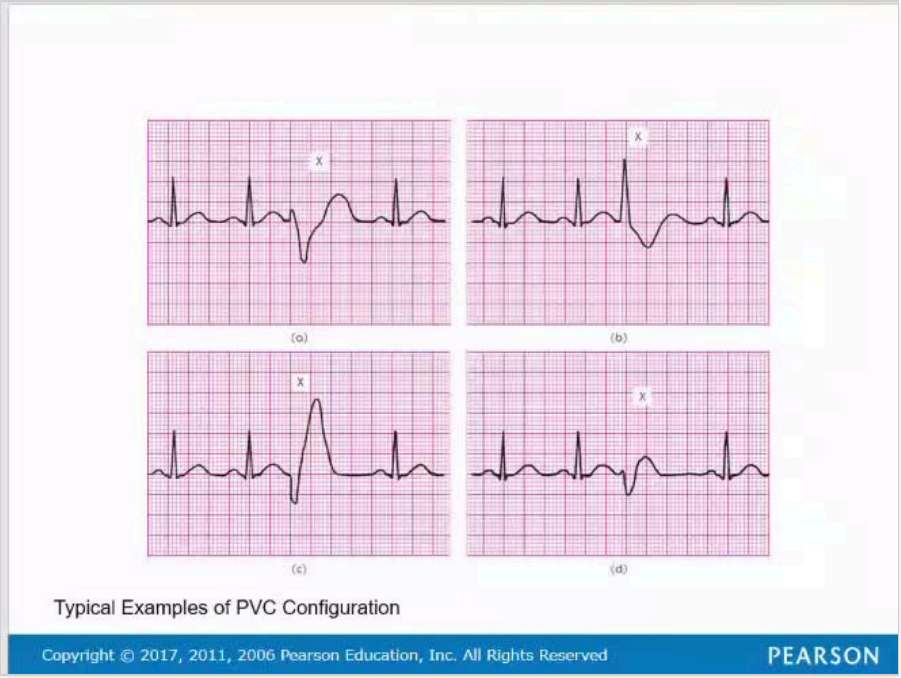
slower, depolarization takes longer, and when we record that on paper traveling from left to right, because depolarization is slower, the QRS is going to get drawn on the paper as a wider complex, 0.12 seconds or greater.

0:05:05 So that's what we're seeing here, is in this complex, it is wide. So there's no P wave followed by a wide QRS, and then typically the other thing that we'll see with these PVCs is the R wave and the T wave go in opposite directions. So here we have a negative deflection for the Q and the R, so the R is negatively deflected, and then we come back up to the isoelectric line, and then we get the T wave. And then typically we get a compensatory pause after a premature beat, which is why you're seeing a little bit of a delay here, and then we go back to an underlying beat from whatever the rhythm is that's pacing the heart, most likely the SA node in this particular case.

0:05:54 And then in example B, again, you see we have the first beat and the second beat are from the underlying pacemaker, most likely the SA node, followed by a beat that occurs early. There's no P wave, and then what we have is a positively deflected R wave in this particular case. We come back down, the QRS is wide, 0.12 seconds or greater, and then the T wave is negatively deflected, so R and T going in the opposite directions. So generally what we're going to see with these beats is an early beat, no P wave, a wide QRS, and R wave and T wave usually travel in opposite directions.

4 Typical Examples of PVC Configuration

0:06:46 – 0:09:17



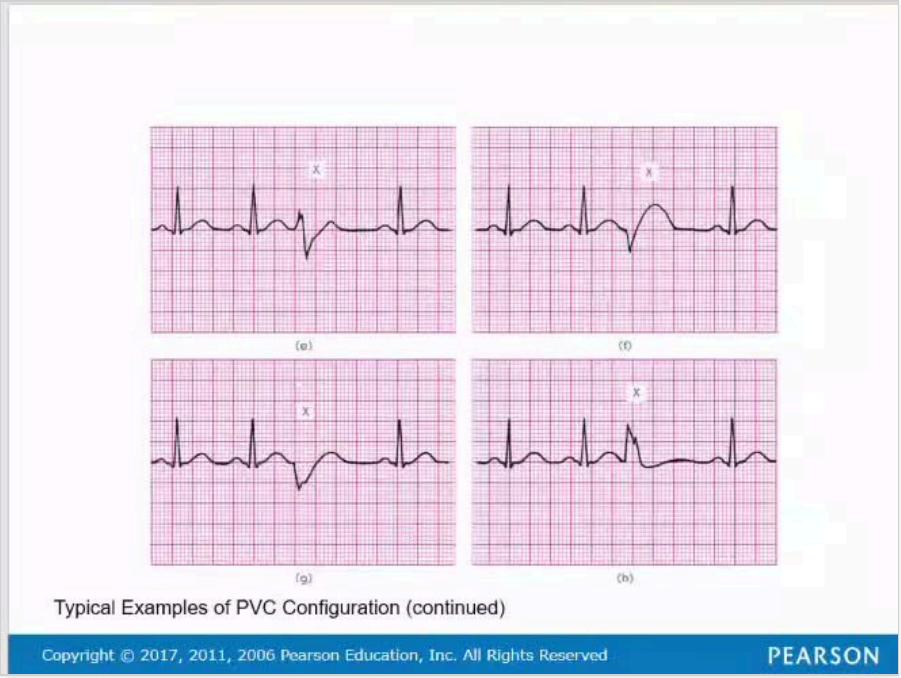
0:06:47 We go to the next slide here and we see again multiple different configurations of these PVCs. So in slide A, small a here, again we have the first two beats are from the underlying rhythm, and then we have this weird looking beat which is, there's no P wave, this is not a P wave, this is actually part of the QRS, so we have this really wide and bizarre QRS, and we're going to talk about how to measure these QRSs as we move along in this lecture today, and then R wave is down, T wave is up, and there's not really a long compensatory pause after this beat, but that's not super important anyways.

0:07:33 I'm just trying to show you that PVCs, depending on where they come from and where the electrical activity is traveling towards, what electrode it's traveling towards, is it traveling towards the right arm, the left arm, is it traveling towards the left leg or the right leg, okay? So we're going to get multiple different configurations, and remember rule of electricity is flowing towards a positive electrode, positive deflection, flowing towards a negative electrode, negative deflection, and sometimes we get flow going towards a positive and negative electrode within the same beat, which is why we'll get some negative deflections and then some positive deflections. Not super important, just trying to explain why we see some positive, some negative within the same beat. In the slide, in the picture B here, again we have two underlying beats followed by a PVC, remember this is a T wave from the previous beat, and then we go right into this early beat, so you can see the spacing here, this is an early beat, we have QRS followed by a

0:08:46 negatively deflected T wave. Same thing here in C, we have an early beat with a wide QRS, R is down, T is up, and then in slide D here, again we have a early beat, R wave down, T wave up, and then we go back into the underlying rhythm. So there's multiple different ways that these PVCs can appear. Again, we'll cover the rules again, I have a slide coming up soon that's going to talk about the different rules. So again, just more examples, E, F,

5 Typical Examples of PVC Configuration (continued)

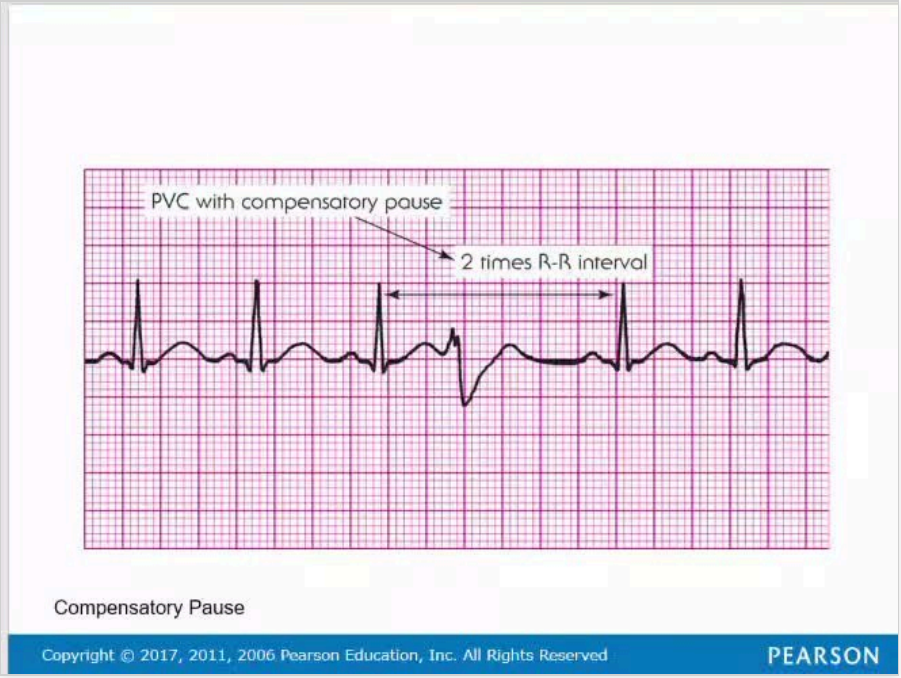
0:09:17 – 0:10:12



0:09:23 G, and H, of how these PVCs can appear. So you can see multiple different views, right, and I don't really care whether it's positively deflected or negatively deflected, what I care about is that you recognize that there's, you know, what are the rules here, and how do I recognize an early beat? Well, an early beat comes close to the previous beat, so it breaks the underlying rhythm, right, so the underlying rhythm, the spacing is regular, and then all of a sudden we have an early beat, and that early beat does not have a P wave, it's a wide QRS, and R and T are usually going in the opposite directions, and you see that again here, for the most part, in E, F, G, and H.

6 Compensatory Pause

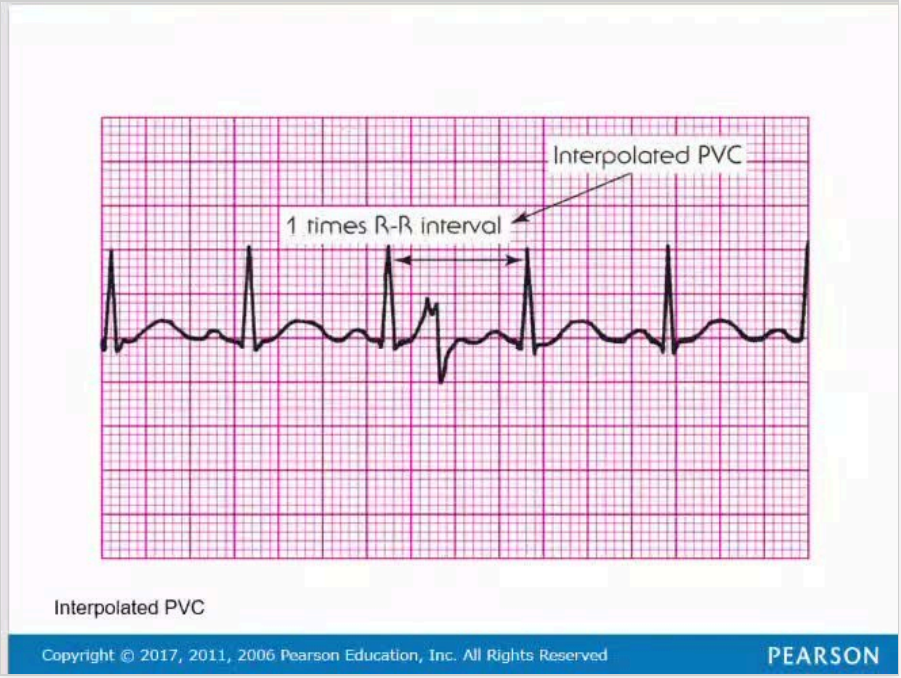
0:10:12 – 0:10:45



0:10:12 The next slide here is going to talk about PVC with a compensatory pause, and that is typically what happens, and usually that pauses two times the R to R interval when you measure it out, not something I'm really going to test you on, not something I would spend a lot of time on, just understand that typically we have a pause after a PVC, and then the underlying rhythm usually kicks back in, and then of course we can, you know, to

7 Interpolated PVC

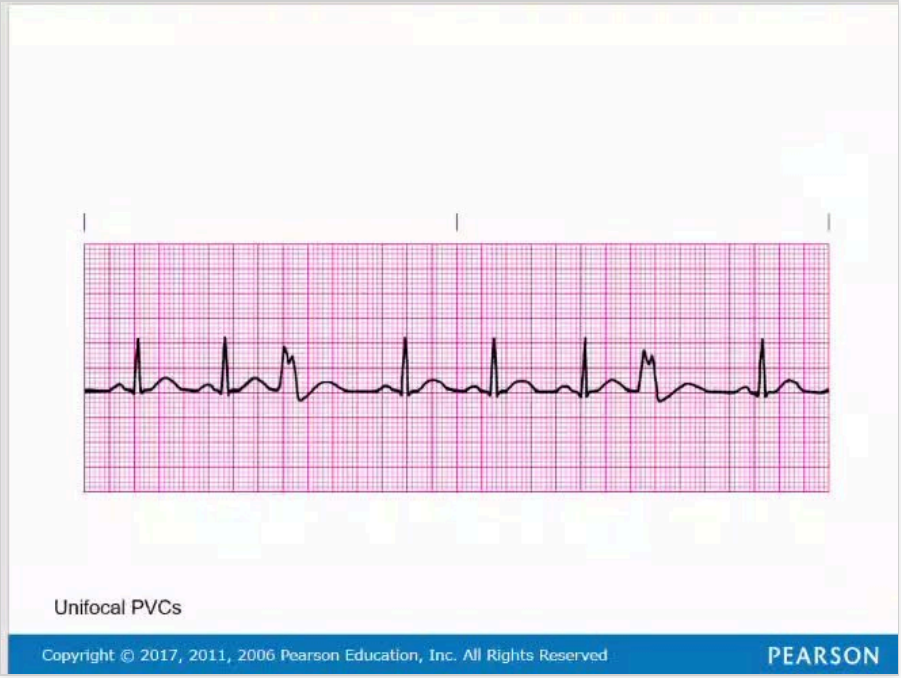
0:10:45 – 0:11:14



0:10:48 break all the rules here, we can have PVCs without a compensatory pause after, and you can see that here, and that's called an interpolated PVC. Again, not something I'm going to expect you to understand in a quiz or a test, it's more important for me that you recognize that that beat is a PVC. It's an early beat, and it's a PVC. Again, we

8 Unifocal PVCs

0:11:14 – 0:19:51



0:11:15 can have multiple patterns here, so when we look at this particular rhythm, we actually can see two early beats. The third beat is an early beat, you can see it's very close to the preceding beat's T wave, and the seventh beat, which looks exactly like the third beat. So in this six second strip, we actually have two PVCs, and it's possible to have multiple PVCs within the same strip, and here I'm giving you two, but it's in a strip. So what's important is, again, I'm going to restate that you recognize that it's a PVC, right? We still have to interpret the underlying rhythm, but we have to also identify the early beats, and then identify where are the early beats coming from, right? Superventricular beats coming from above the ventricle have narrow QRSs, premature ventricular complexes have

0:12:31 wide complexes. So I'd like to actually take a moment here and gather information for this particular strip. So I'm going to go through and show you how I would interpret this strip here. So I'm going to use my cursor here, and the first thing I'm going to do is gather my five pieces of information. So the first thing I want to know is, is this strip regular? So I'm going to do 10, and 3 make 13, and 4 make 17. And it's possible that this second beat is halfway in between, and that's halfway in between, so this spacing could be anywhere from 17, or 17 and a half, or 18 small boxes between these first two beats.

0:13:27 So typically what I would do is, on a piece of paper, write down, you know, we'll split the difference, and we'll go 17 and a half between these two beats. 5, 10, 11, and a half. So 11 and a half between the second and third beat. 5, 10, 15, 20, 24 and a half between the third and the

fourth, and then we go back to 5, 10, 15, 16, 17, and I'm going to call it two halves and make that 18. So we have 17 and a half here, we have 18 here. 5, 10, 15, 16, 17, 18, 18, 5, 6, 7 and a half, 8, 9, 10, 11, 5, 10, 15, 20, 21, 22, 23 and a half. Okay, so this rhythm is actually regular because the spacing from these two beats, and the spacing from these two beats, and these two beats are all within a box and a half. So the underlying rhythm here is regular, so for the regularity I would put it is regular,

0:15:06 and then there are two early beats. So I typically, for the regularity, I would say this rhythm is regular with two ectopic beats. Now I got to do the heart rate. I'm going to say my spacing here is 18 small boxes, so 1500 divided by 18 gives me a heart rate of 83. The next thing I want to know, P waves, and when I look at my P waves, these P waves, the first, the second, the fourth, fifth, sixth and eighth, all of those P waves look the same, and they are positively deflected, so they're upright, and they all look the same, so I would call them uniform. So the fourth piece of information here is going to be my PRI, and this P wave starts right here,

0:16:07 and my Q wave dips right there. So this is going to be four or four and a half small boxes, so my PRI is going to be 0.16 or 0.18. I actually think it is four small boxes, so I'm going to count between, so I'm going to count moving left to right, one, two, three, four. This third beat does not have a P wave, and it's early, it breaks up my spacing, so I know this is a premature beat here without a P wave, and this QRS looks pretty wide. So I'm going to continue counting my PRI for the underlying beats, one, two, three, four, one, two, sorry, one, two, three, four, and then that one dips there, four. So my PRI is consistently 0.1, I'm sorry, it is four small boxes, so my PRI is going to be consistently 0.16 seconds. Now I'm going to measure my QRS, the underlying QRS, one, two, one, two, one, two, so I'm pretty confident that my underlying QRS

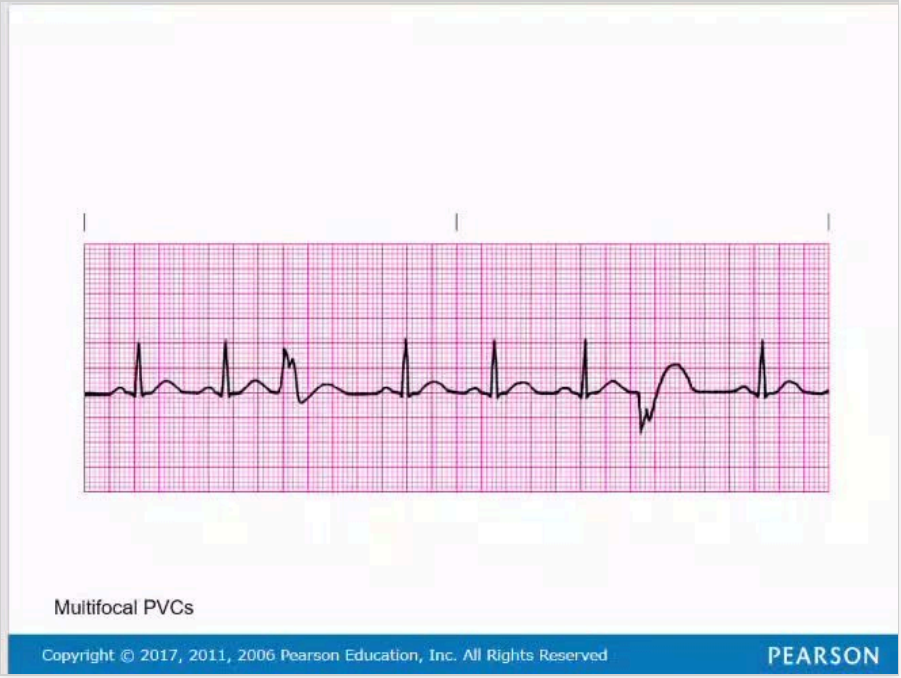
0:17:37 is 0.08 because it's two small boxes, that's a narrow QRS, so I know these are superventricular beats, so I have the underlying rhythm which is regular, so everything is normal, the heart rate is normal, it's regular, the heart rate is normal, the P waves are upright in uniform, I have a normal PRI and a normal QRS, so this underlying rhythm can only be coming from the SA note. Because the underlying rhythm, normal sinus rhythm, now I need to investigate where these two premature beats are coming from, so there's no P wave in front of either one of them, they're early, and if I measure this QRS, it starts right there and it's going to end right there, so one, two, three, four,

0:18:43 one, two, three, four, so the underlying, the QRS for these early beats is going to be 0.16 seconds, so I have no P wave with the wide QRS and they're early beats, so these have to be PVCs, and these PVCs, if I cut this one out and put it over the top of this one, they are exactly the same, so when these early beats look the same, it means they're coming from the same irritable focal point, so when they come from the same focal point and they look the same, we're going to call them unifocal PVCs, so my full interpretation here is going to be normal

sinus rhythm with two unifocal PVCs, so that's a lot to put together, it's going to take a little time and a little work on your part to be able to put this together, but I went through exactly how we should do that, so now I'm going to move on to the next slide, and now we have the same underlying

9 Multifocal PVCs

0:19:51 – 0:21:45

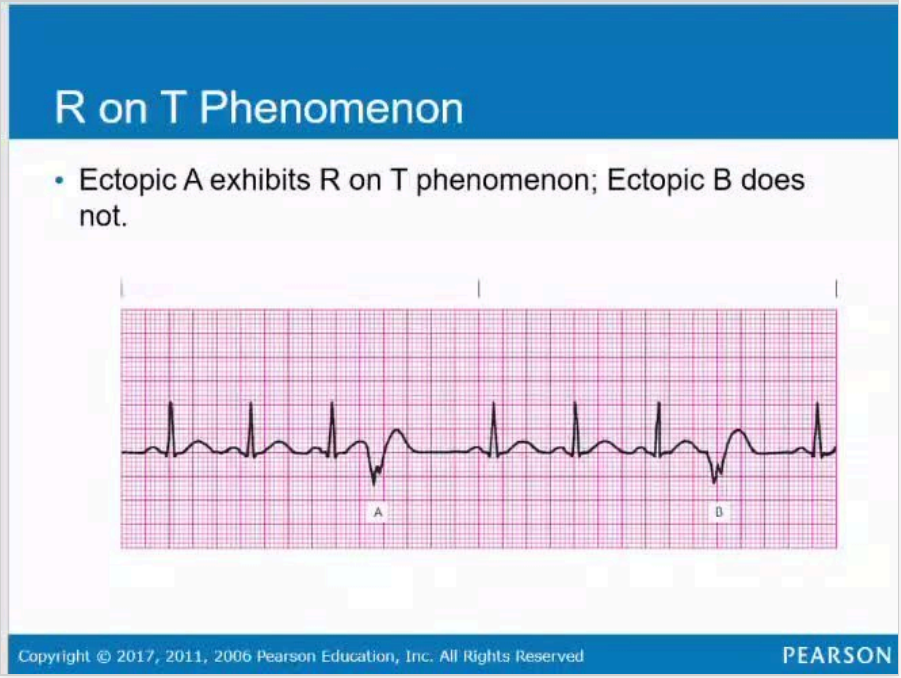


0:19:54 rhythm, and we have two PVCs in this, but those PVCs look different, so as I go through my information gathering here, the spacing for my regularity, this is going to be 18, 17 and a half or 18, this is short because this is a premature beat, and it's wide with no P wave, go back to 18, 18, and then this is early, so I know that's a premature beat, no P wave with a wide QRS, so I already have an idea that both of these early beats are coming from the ventricle, and then I go back to the underlying rhythm, so this heart rate, this rhythm is going to be regular with two ectopic beats, the heart rate is going to be 83, the P waves are upright in uniform, this is the same exact rhythm as the previous one that I went to, the only thing that's different here is that the PVCs look different, so the PRI is still 0.16, the underlying QRS is still 0.08,

0:21:01 the width of these QRS is still four small boxes, except now they look different which means there's more than one irritable focal point, so there's an irritable focal point for this one and a different irritable focal point for this one, so because they look different we're going to now call these multifocal PVCs, so as I put this whole thing together I'm going to have normal sinus rhythm with two multifocal PVCs, premature ventricular complexes. All right moving on to the next slide here, not something that I'm

10 R on T Phenomenon

0:21:45 – 0:23:40

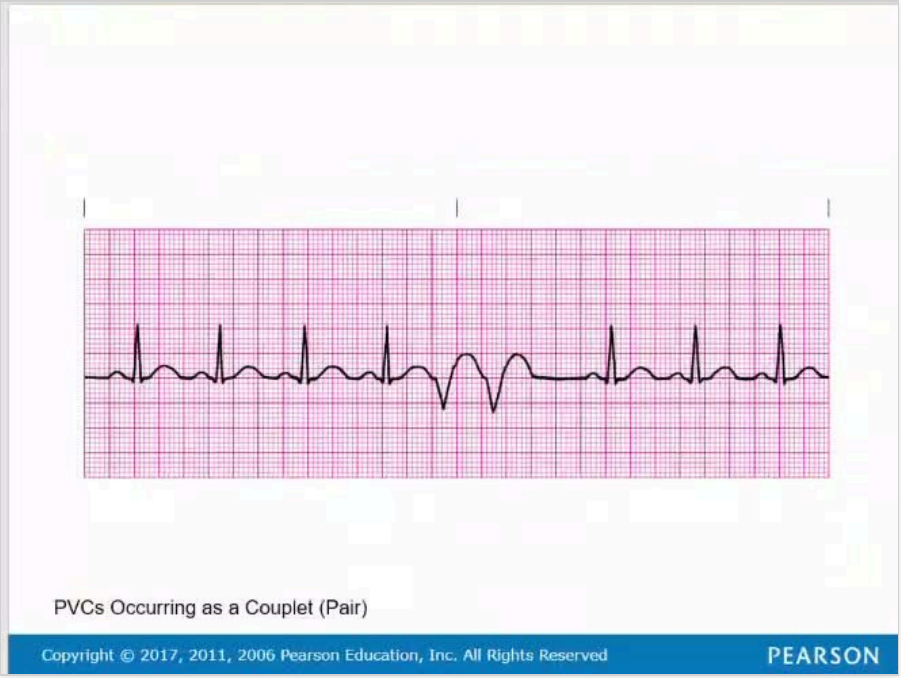


0:21:49 going to test you on in this particular slide, but something that's important to keep in the back of your head because this phenomenon is potentially life-threatening, so it's called R on T phenomenon, so when we look at the first PVC in this rhythm string, what this R on T means is this is the T wave from the preceding beat which connects right to the R wave, so that's why we call it R on T, and the reason why this is potentially life-threatening is if you remember this beat is occurring during the repolarization phase of the ventricle because it's right on the T wave, T wave represents ventricular repolarization, so when we get an impulse strong enough to depolarize during that time frame, what's dangerous about that is it could trick the system into a life-threatening arrhythmia like ventricular fibrillation, which we're going to talk about in this lecture, or ventricular tachycardia, which we're going to talk about in this lecture as well, so this is potentially

0:23:11 life-threatening because it could flip the patient's heart rhythm into a life-threatening rhythm, and then when we move down here to B, there is a small space here at the isoelectric line before this beat actually occurs, so there's no R on T phenomenon for the second premature ventricular complex in this rhythm. Moving on to

11 PVCs Occurring as a Couplet (Pair)

0:23:40 – 0:26:11

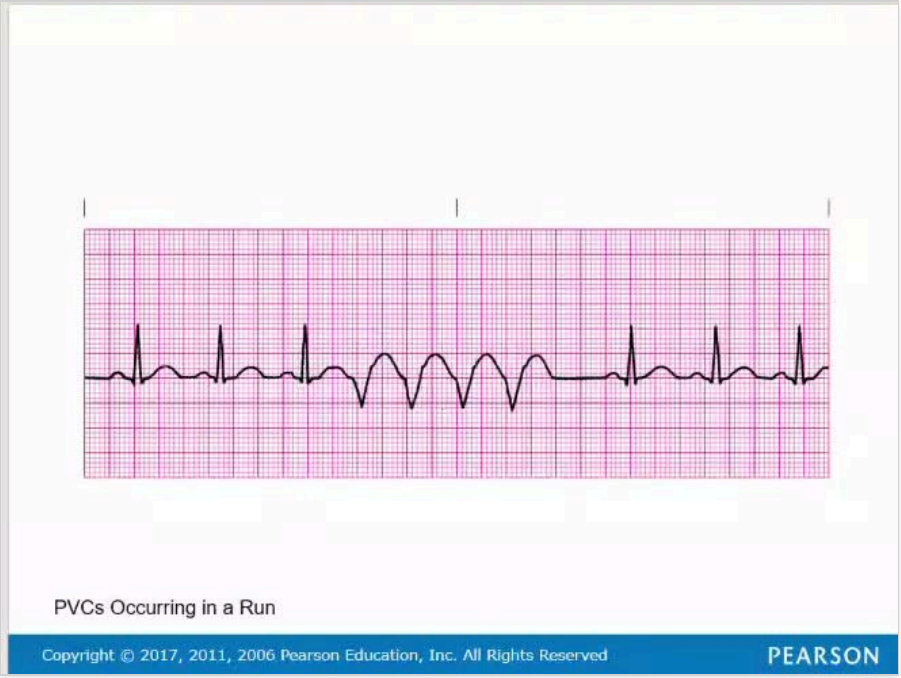


0:23:41 the next slide, again we're covering all of the different patterns that premature ventricular complexes can have, and in this particular case you can see that I have two premature ventricular complexes in a row, so this is the same underlying rhythm as the previous ones, the spacing here is going to be 18 small boxes, so 18, 18, 18, this is going to be less than 18, this is going to be probably somewhere around 9, maybe 10, I have a compensatory pause, and then I go back to 18, so the underlying rhythm here is regular interrupted by two premature beats, so it's regular with two ectopic beats, the heart rate is still going to be 83, P waves are going to be upright in uniform in the underlying rhythm, there's no P waves for the premature beats, this PRI is still going to be four small boxes across the strip, so my PRI is 0.16, the QRS here are two small boxes, so that's going to be 0.08, the PVCs as we measure this out

0:25:04 are going to be one, two, three small boxes, they're both going to be three small boxes, but there's no P wave, R and T are in opposite directions, so both of these are premature ventricular complexes, when we see two in a row we call it a couplet, like a couple holding hands, right, they go back to back, so my total interpretation here is normal sinus rhythm with one couplet of PVCs, again some work to put this together, so you're going to have to be able to recognize all of these different patterns, like unifocal PVCs, I might give you a rhythm with one PVC in it, or two PVCs, and those PVCs look the same, they're unifocal, or I give you a rhythm with two different looking PVCs, that's multifocal, or I give you a rhythm with two PVCs in a row, that's a couplet, and then as we continue to move on here, now you're seeing a

12 PVCs Occurring in a Run

0:26:11 – 0:28:28

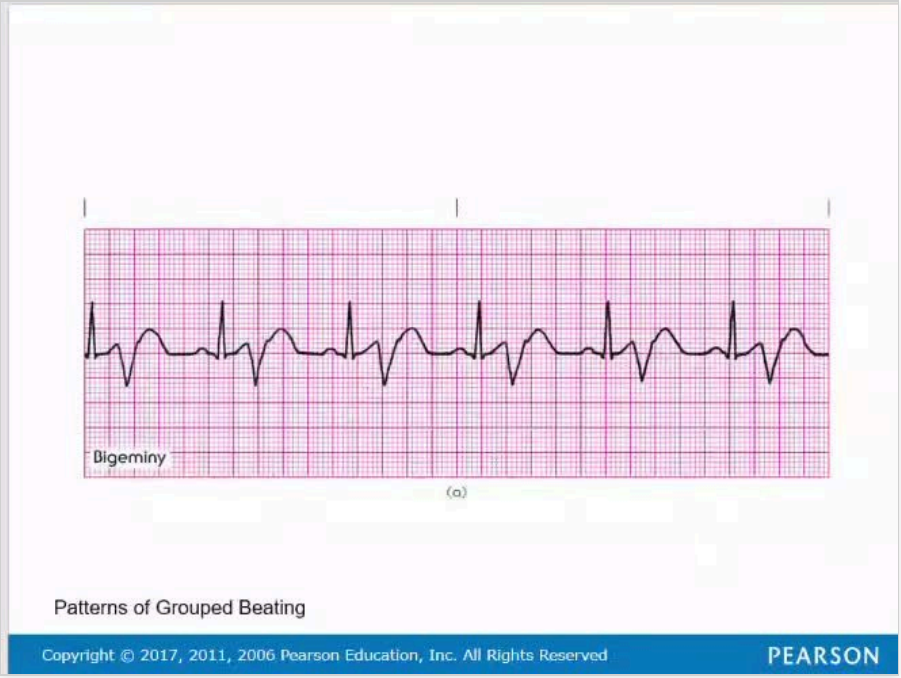


0:26:13 rhythm, this is again the same rhythm, but now I'm seeing four premature beats in a row, so these are four PVCs in a row, so this underlying rhythm is regular 18 18, less than 18, this is probably 10 10 10, a longer space because I have my compensatory pause 18 18, so the underlying rhythm is regular, interrupted by four ectopic beats, heart rate is still going to be 83, P waves are upright and uniform, PRI is four small boxes, 0.16, underlying QRSs are two small boxes at 0.08, and then the PVCs are going to be three and a half small boxes, you see it starts right there's the negative dip right at that line, and then it stops with this little hitch right here halfway in between, so I have one two three and a half small boxes, so these PVCs are going to be 0.14 seconds, so they are wide, they're early, no P waves, and they're wide, that's how I know so any time we get three, one two three or more,

0:27:46 so three or more PVCs in a row, we're going to call it a run or a burst of VTAC, otherwise known as ventricular tachycardia, so in this case we have a burst, so my overall interpretation for this rhythm would be normal sinus rhythm with a four beat run of VTAC, again it's going to take some time for you to put this together, okay so we're not

13 Patterns of Grouped Beating: Bigeminy

0:28:28 – 0:34:59



0:28:30 done with PVCs and patterns yet, so if you look at this rhythm, this rhythm is kind of confusing, so I want to orient you, I could see a narrow QRS for this first beat, I can't see what happened before this QRS, and then I have an early beat, so I have a beat coming right off of the T wave, so again I'm not going to test you on this, but this is R on T right here, so I have a negative R wave followed by a positive T wave, there's no, this is not a P wave, this is the half, the first half of the preceding beats T wave, then I go into a little bit of a pause, and then I have a P wave followed by a QRS, followed by half of a T wave, and then into another beat R on T, which is going to be wide without a P wave, and then R and T going in opposite directions, and this pattern repeats

0:29:36 itself across the rest of the strip, so I have a regular underlying beat followed by a PVC, a regular underlying beat followed by a PVC, and that repeats itself all the way across the strip,

0:29:53 so I have a PVC every other beat, so when I see a PVC every other beat, we're going to call it gather all of our information so that you can understand how to put this rhythm together with a final interpretation, so I'm going to count out the spacing between the underlying beat to see if the rhythm is regular or not, five, ten, fifteen, twenty, twenty-one, twenty-two, and a half, that's a half, and then another half makes 26, so there's 26 small boxes between the first and the second underlying rhythms, R waves, five, ten, fifteen, twenty, twenty-one, twenty-two, twenty-three, and a half, twenty-four and a half, twenty-five and a half, and a half makes 26, five, ten, fifteen, twenty,

0:31:08 twenty-four, and a half, twenty-five and a half, twenty-six, so R to R here, this R to R spacing for the underlying QRSs is 26 boxes, and so that means that the underlying rhythm is regular, there happens to be an ectopic beat here, here, here, so there's an ectopic beat in between the underlying rhythm's beats, so we're going to say, because this spacing is the same, that the underlying rhythm is regular, and then there's an ectopic beat in between, so the regularity for this rhythm is going to be regular with an ectopic beat every other.

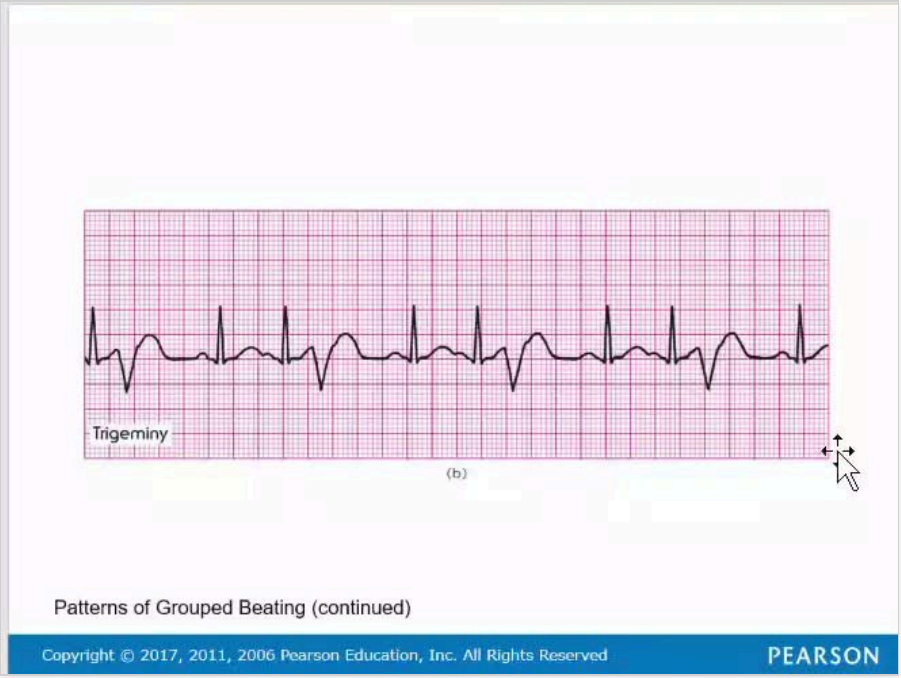
0:32:05 Our heart rate is going to be determined by the spacing of the R waves for the regular rhythm, because it's, because it's regular, so fifteen hundred divided by 26 was our spacing, gives me a heart rate of 57.6, so little red flags throws up in the back of my head, this heart rate is less than 60, so the underlying rhythm is bradycardic, so I have to keep that in mind as I go forward towards my interpretation. So the underlying rhythm's P waves are upright, and they look the same, the ectopic beats in between don't have P waves, so for the underlying rhythm the P waves are upright and uniform, and one, two, three, four, so again this is the same rhythm, so this PRI for the underlying rhythm is going to be four small boxes across the strip,

0:33:14 making it 0.16. QRS, again this is the same QRS, these are two small boxes, so the underlying rhythm has narrow, normal QRSs, and then our ectopic beats are going to be one, two, three, and a half. So the QRS for the underlying rhythm at three and a half is going to be 0.14, so because there's no P wave, these QRSs are wide, R and T are in the opposite direction, they have to be premature ventricular complexes, and in this particular case they're appearing in a pattern, and they are interpretation together, we're going to interpret this rhythm because the overall heart rate's less than 60, we don't count the premature beats in the heart rate, our heart rate was 56 or 57.6, so the underlying rhythm is going to be sinus bradycardia, sinus bradycardia because the heart rate was under 60, P waves upright and uniform, normal PRI, normal QRS, that's what makes it coming from the sinus node, so sinus bradycardia, and then PVCs, every other beat makes it

0:34:42 bi-gemini, so this is going to be sinus bradycardia with bi-gemini, again a lot to put together, so you guys are going to have to do some work on this, and so if we can have a PVC appear every other beat, we can have a PVC appear every third beat, so you're going

14 Patterns of Grouped Beating: Trigeminy

0:34:59 – 0:39:39



0:35:03 to see a repeating pattern here, it is an underlying beat, an underlying beat followed by a PVC, underlying beat, underlying beat followed by a PVC, underlying beat, underlying beat followed by a PVC, all right so let's go through all of our information gathering again, so you know how to piece this together, so again I want to count the spacing between our waves, so let's do that 5, 10, 11, 12, and it looks like that's halfway in between, and that's halfway in between, so we're going to call this 13, this is going to be one, that's a half, one and a half, two and a half,

0:35:59 three and a half, four and a half, five and a half, six and a half, and a half makes seven, so there's seven here, I know this is going to be longer because I have my compensatory pause,

0:36:11 five and four make nine, 10, 11, 12, 13, so I have 13, 13, and this is going to be 13, so I have, remember the rule of thumb, if I can find in the underlying rhythm, if I can find two r to r intervals that are within a box and a half, that's going to make the underlying rhythm regular, that's an important rule to keep in mind, so even though this rhythm is scattered, when we get two beats together from the underlying rhythm, there is consistent spacing, so that makes this rhythm regular, regular with an ectopic beat every third, so now my heart rate is going to be based on the spacing of the underlying rhythm, so we're going to go 1500 divided by 13, gives me a heart rate of 115, so again that's something I got to carry forward with me because that heart rate is above 100, the underlying rhythm here is tachycardic, P waves for the underlying rhythm are upright and uniform, we can only

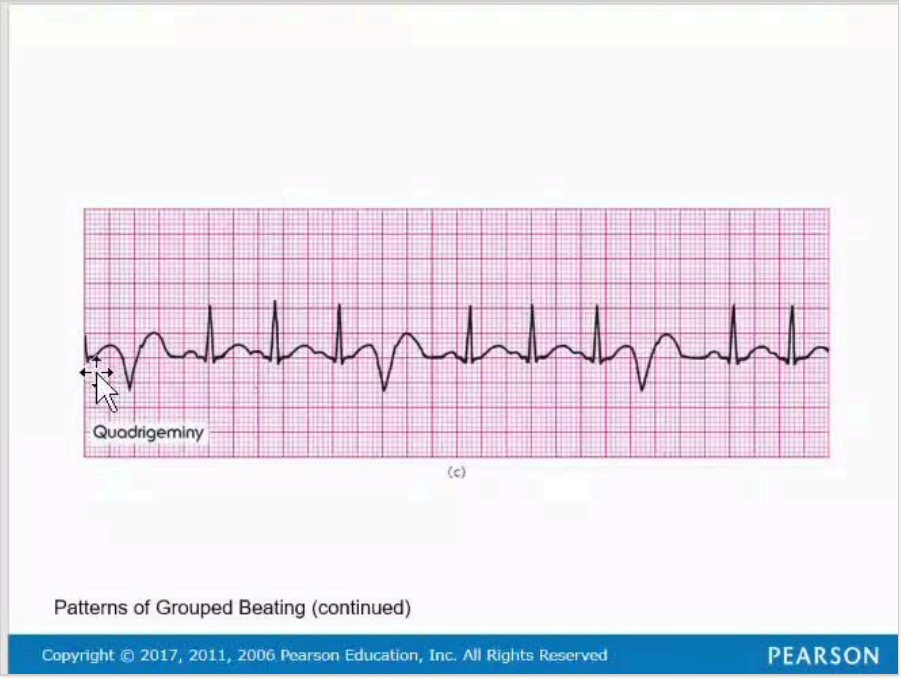
0:37:35 see a part of this because the heart rate is fast enough, the space gets smaller between beats, but I can see enough of this P wave here that it looks dome shaped just like that one, so there's a P wave that's upright and uniform for the underlying rhythm, and this PRI is going to be four small boxes at 0.16, there's no P waves for the ectopic beats, again this is an R on T phenomenon,

0:38:10 and then when I measure my QRS's they're going to be two small boxes, pretty consistently that's going to be 0.08, QRS's for these ectopic beats are going to be, this one actually looks like it's it starts right there, so that's one two three, and then halfway in between there, so my QRS's for these premature beats is going to be three and a half or 0.14 seconds, so because I have no P wave, a wide QRS, R is negative, T is positive, I know that these are PVC's or premature ventricular complexes that occur every third beat, so this pattern we're going to call trigeminy, so my full interpretation here is because the underlying rhythm's heart rate is at 113, we're going to call this sinus tachycardia, it's coming from the sinus because it's regular, P waves look the same, normal PRI narrow QRS, everything is normal, can only be coming from the SA node, because it's above 100 that's what makes it sinus tachycardia, so we have sinus tachycardia

0:39:27 with a PVC every third beat which makes it trigeminy, so this is sinus tachycardia with trigeminy, and then the next rhythm we're going to talk about

15 Patterns of Grouped Beating: Quadrigeminy

0:39:39 – 0:42:43



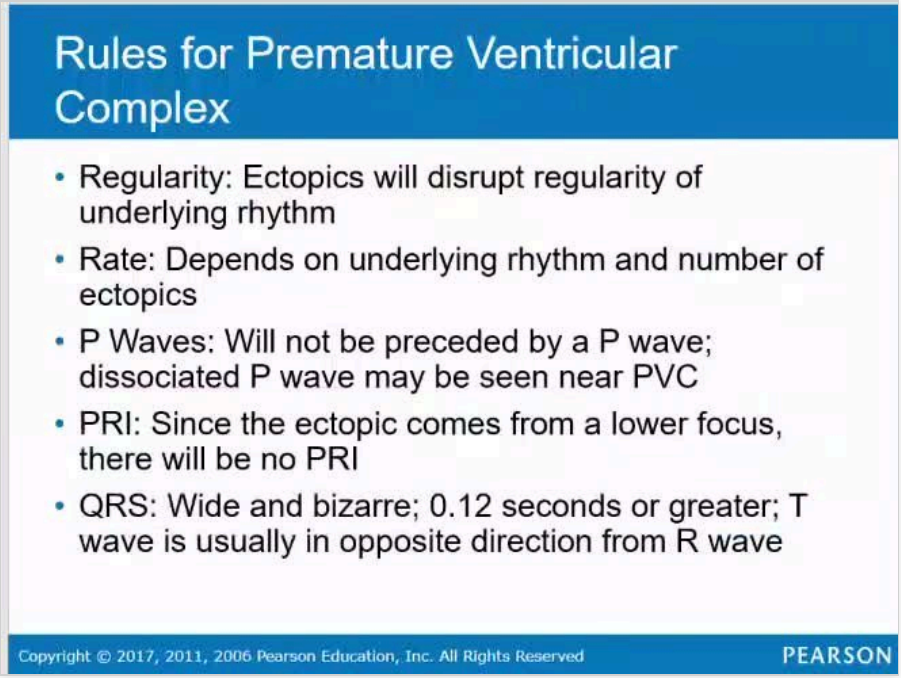
0:39:40 is a PVC every fourth beat, so we can't see what happens over here, but we have an underlying beat, followed by an underlying beat, followed by an underlying beat, followed by a premature beat, again R on T phenomenon, no P wave, wide QRS, R and T going in the opposite directions, and then we have a pattern that repeats itself, one, two, three underlying beats, followed by a PVC, so we're going to gather our information again, and this spacing still looks like it's 13, so this is the same heart rate for the previous rhythm, so the spacing for the underlying beats is going to be 13, and I see one, two, three, four, five different spots where the spacing for R to R is 13, that makes this underlying rhythm regular, and then I have a PVC every fourth beat, so the rhythm is going to be regular with an ectopic every fourth beat, my heart rate is going to be 113, 115, excuse me, 1500 divided by 13 gives me

0:40:55 115 is my heart rate, P waves are upright in uniform, and the PRI is four small boxes at 0.16, the none of the premature beats have P waves, my underlying QRS is two small boxes, one, two, so my QRS is 0.08, narrow, and then my QRS for these ectopic beats again are going to be three and a half, which makes the premature ventricular complex QRS measurements at 0.14 seconds, and now I have to put my total interpretation together, sinus tachycardia with quadrigeminy, which means I have a PVC every fourth beat, so a lot to cover with all of these different PVC patterns that can occur, so you know in the quiz, you know I'm going to try to give you a smattering of all of these different patterns, and you're going to be able to have to appropriately, so you need to do some work over this next week because these are the challenging ones, and these are going to be more of the life-threatening ones as well,

0:42:20 so you know when you're when you are monitoring patients during the night while you're doing sleep studies on them, you know none of these rhythms that I've covered so far are life-threatening, but the ones that I'm going to get into next are life-threatening, so these are going to be important, this chapter is super important, okay so rules, and we've been talking about the

16 Rules for Premature Ventricular Complex

0:42:43 – 0:44:51



0:42:47 rules for these premature ventricular complexes as I've been going along, so the regularity for these ectopic beats is most is really going to depend on the underlying rhythm, and as you saw the underlying rhythm we're looking at R to R spacing, these beats are going to be early, so they're actually going to disrupt the regularity of the underlying rhythm,

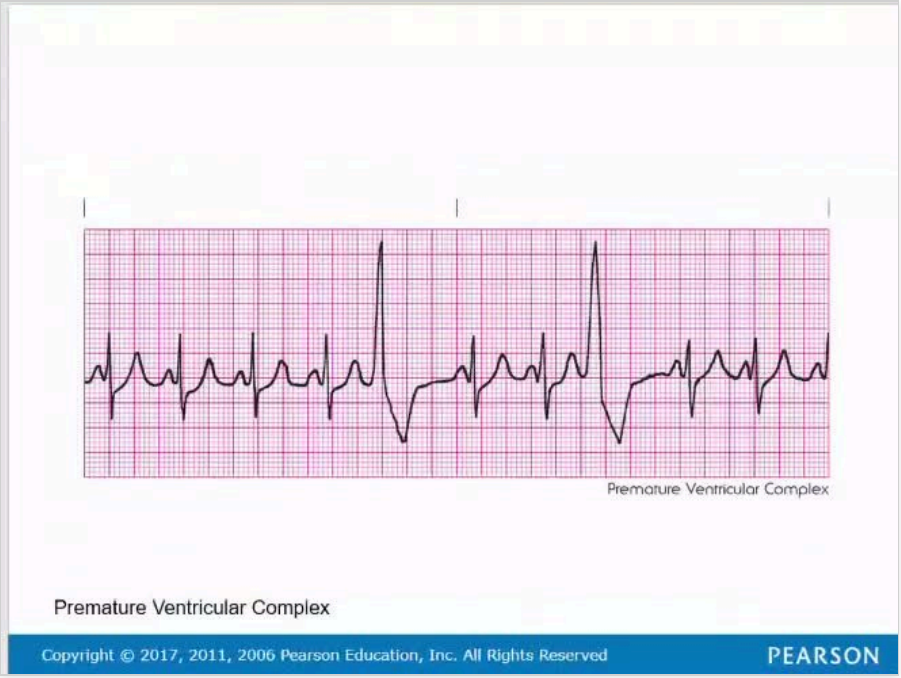
0:43:19 so you might see PVCs that come that are visible in a rhythm strip where the underlying rhythm is regular, then you're going to count the R waves between those regular portions at your heart rate, but I might also give you a rhythm where the PVCs are in an underlying rhythm that is irregular, so you're going to also have to remember when it's an irregular rhythm we don't count the PVCs, but you would count the number of narrow complexes in that six second strip, multiply it by 10, and that's going to be your heart rate,

0:43:55 so again the rate is going to depend on whether the rhythm is regular or irregular, you should know the two methods that we use to count heart rate at this point, PVCs will not have a P wave in front of them, I talked about that several times as I went through the different patterns,

0:44:16 because there's no P wave there will not be a PRI, but you still need to assess P waves for the underlying rhythm and the PRI for the underlying rhythm, the QRS for these premature ventricular complexes will be wide and bizarre, they'll be 0.12 seconds or greater, T wave usually goes in the opposite direction of the R wave and I have been talking about that, so this slide is just reinforcing the rules that we were discussing going through all of the different patterns that PVCs can appear, so this is just one more slide

17 Premature Ventricular Complex (ECG strip)

0:44:51 – 0:51:11



0:44:57 here giving you an example of some PVCs that may appear, so you know I'm going to take the time again to go through the information gatherings, the more I could do this the more you guys will be able to try to put this whole thing together, so the first thing we're going to do is assess our regularity, you can see that you know we have a P wave followed by a QRS T wave, P wave, QRS T and that pattern continues until I get to this point here where I have an early beat, so if you look at the spacing between the underlying beats there's you know there's maybe three or four small boxes before the next beat occurs, I don't have that spacing from here to here, so this is an early beat, there's no P wave associated with it, R wave is up here, T wave is down there, it looks like it's going to be at least three small boxes, so I'm going to go ahead and assume that that is a premature ventricular

0:46:02 complex, then we go with a compensatory pause back to the underlying rhythm and then I have a similar wide complex beat and then back to the underlying rhythm, so I'm going to start by counting my spacing here, this looks like 14 boxes, this is 14, this is 14 and again I'm going from this R wave to this R wave, 14 small boxes between and then here I'm going to have 5, 10, 11 and a half between these two and then I'm going to have an elongated space because of this compensatory pause and then I'm going to get back to 14, this is going to be somewhere around 11, long again because of the pause but then I'm going to go back to 13 and a half or 14 and then 14 again, so regular, regular, regular, this is an early beat, regular, an early beat, regular, regular, so I have two ectopic beats, so my regularity here is going to be regular with two ectopic beats, I'm going to calculate out my

0:47:18 heart rate using the divide by 1500 method because the spacing is regular, so 1500 divided by 14 equals 107 is the heart rate for the underlying rhythm, the underlying rhythm has P waves that are upright and look the same all the way across the strip, the PRI, let's measure this out because this is a new PRI and sometimes I count backwards, I find where my QRS starts and I count backwards, so this is what I'm going to do here, this QRS starts right on that line, right there, so I'm going to count backwards, one, two, three, maybe three and a half, one, two, three, one, two, three, so I'm going to say that this PRI is three small boxes at 0.12, still normal because remember the normal range for the PRI is 0.12 seconds to 0.2 seconds or three to five small boxes, now I'm going to measure my QRS and this QRS, again I'm going to count backwards because that stops right there and it starts right there,

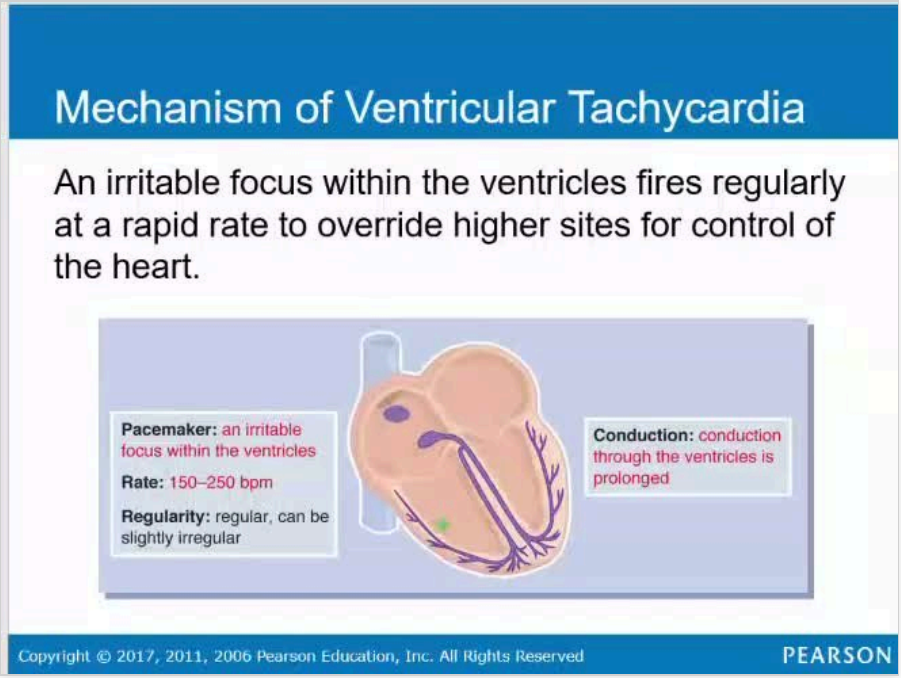
0:48:49 so one, two, one, start halfway there, right there, so I'm going to go halfway, two, so that's two small boxes, so the underlying QRS is 0.08, now I'm going to count the PVC here, starts right there and what you want to do is count until you see this little notch and I think it ends right about there, it's easier to see on the second PVC, you see this bend right there, that's where the PVC ends, it starts here and it ends there, so I'm going to kind of use that same spot right about here, one, two, three and a half,

0:49:49 one, two, three and a half, so the PVCs are three and a half small boxes so that would give me 0.14 seconds in time, so this rhythm is regular with

0:50:07 two premature ventricular complexes, my spacing is 14 so that gave me a heart rate of, I know I did this already but I just want to make sure I'm consistent, 107, the P waves are upright in uniform, the underlying QRSs are 0.08, the ectopic QRSs which don't have a P wave are 0.14, R and T are going in opposite directions and these PVCs look the same, so that means they're coming from the same focal point, so my full interpretation for this particular rhythm is going to be sinus tachycardia because the heart rate is 107 with two PVCs that look the same, so they're going to be unifocal PVCs, so this is sinus tachycardia with two unifocal PVCs, okay we're going to continue on here,

18 Mechanism of Ventricular Tachycardia

0:51:11 – 0:52:26

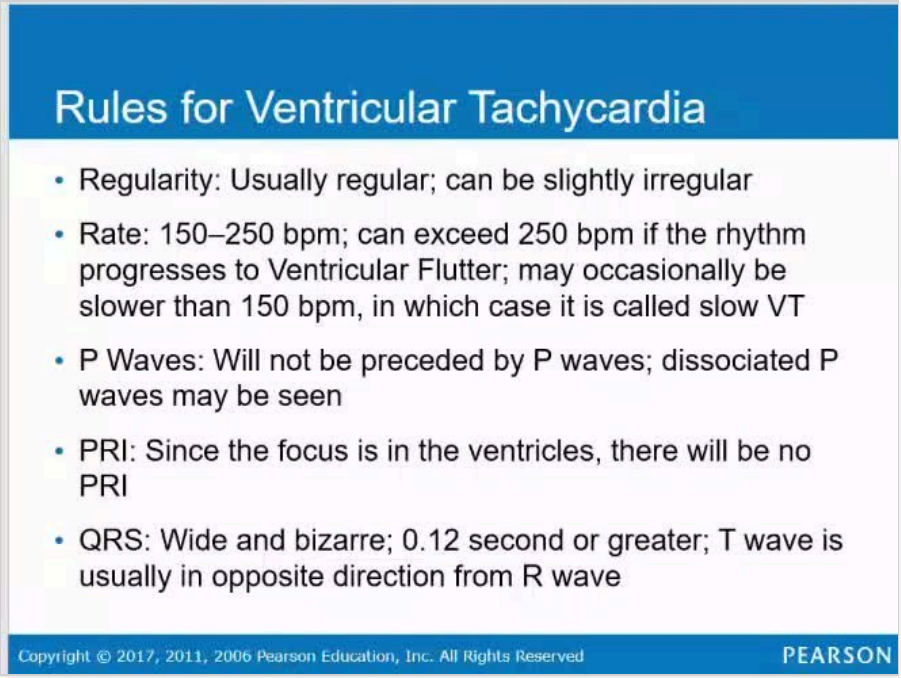


0:51:11 I'm going to move away from PVC patterns and I'm going to discuss ventricular rhythms moving forward, so the next rhythm I'm going to talk about is ventricular tachycardia, so this again is an irritable focal point that now fires but it fires repetitively, not just one single beat to make a PVC, it fires repetitively and it's going to fire repetitively at a rate faster than the SA node can fire, so now it's going to take over as the pacemaker and generally the inherent rate for this ventricular tachycardia or this irritable focal point usually fires at a rate somewhere between 150 to 250 beats per minute,

0:52:07 so we're going to look at the rules and then we're going to interpret, so this is the same irritable focal point that's firing except now it's firing repetitively at a rate higher than the SA node, so now it's going to take over, so all we're going to see here is ventricular beats

19 Rules for Ventricular Tachycardia

0:52:26 – 0:53:54

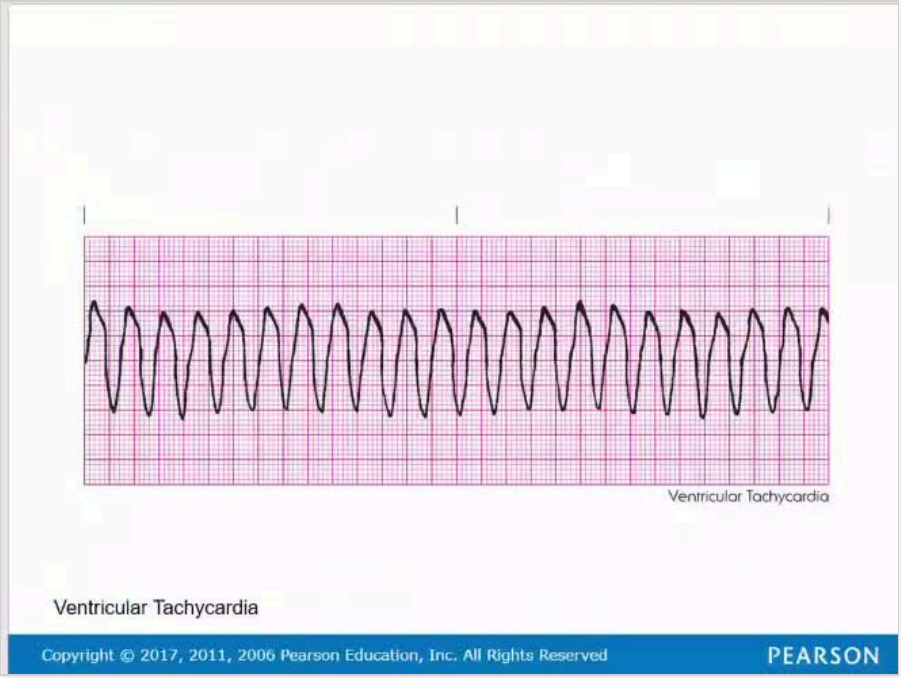


0:52:29 throughout this entire rhythm, so the rhythm itself for ventricular tachycardia is usually regular, there can be some times where it can go slightly irregular but it's usually regular, as I said the rate is typically 150 to 250, it can go faster than 250, if it's above 250 then we actually change what we call it, we would then call it ventricular flutter, I've been interpreting rhythms for about 30-35 years and I do a lot of intensive care clinical and I have actually never seen ventricular flutter, so I'm going to tell you I will not put it on a quiz or I will not put it on a test, you will see ventricular tachycardia but I will not give you ventricular flutter and then occasionally this rhythm's heart rate can be slower than 150, if that's the case then we just call it slow V-tach, there will not be any P-waves, so therefore there will not be a PRI to measure, the QRS will typically be wide 0.12 seconds

0:53:46 or greater and usually R and T go in opposite directions, so that's the rules, let's take a look at the rhythm and let's go through our interpretation, so when

20 Ventricular Tachycardia (ECG strip)

0:53:54 – 0:58:23



0:54:04 you first look at this most people learning how to interpret EKGs kind of have a difficult idea of okay is the R-wave on the positive side or is the R-wave on the negative side, in terms of assessing the regularity I'm not sure it's going to make a difference because the spacing up here is going to be the same as the spacing down here, but I will tell you I've been looking at EKGs for a long period of time the R-waves are down, so the T-waves are up right, so typically the R-wave is going to be a little bit more pointy and not as rounded, these T-waves have a little bit more roundness and these are a little bit more pointed and so the R-waves are going to be negatively deflected and the T-waves are going to be positively deflected, so we're going to just start gathering information like we normally do here if we're going to count our spacing between R-waves, so I have it starts right there so I have four, five, six, seven, two,

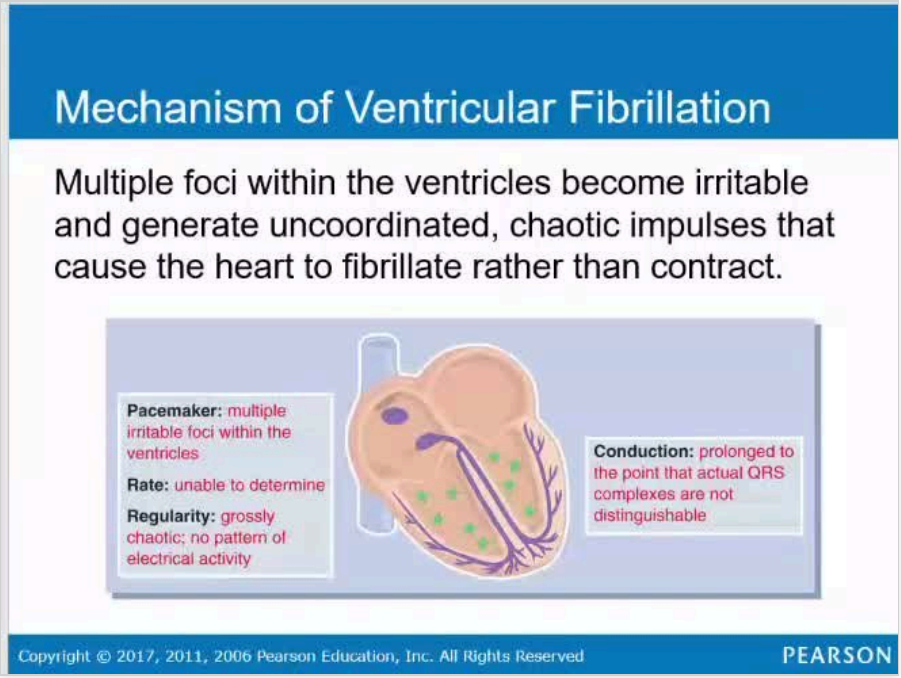
0:55:19 and five makes seven, five, this might be six and a half but it's close enough to seven and this might be six and a half but I think it's close enough to seven, so my spacing R to R here cross this rhythm strip is going to be seven small boxes, so that makes this rhythm regular, so then I'm going to count my heart rate 1500 divided by seven is going to give me a heart rate of 214 beats per minute, there is no P-wave so I can't measure a PRI and now I need to measure my QRS, as I said this is the end of the T-wave, see this little notch here is the beginning of the preceding beats T-wave and the T-wave ends here, so this is where my Q-wave, R-wave, S-wave, you see that little bend right there, so I'm going to measure from here to here and it looks like it's right there three, maybe three and a half, close enough, so we're going to say 0.12 seconds for our

0:56:49 QRS width, so I have nothing but wide and bizarre QRS complexes with no P-waves, R and T are going in the opposite direction, this has to be ventricular tachycardia, can only be ventricular tachycardia and typically this rhythm occurs without a pulse, so this is immediately life threatening, so if you're doing a sleep study and you see this rhythm pop up on your screen, first thing you're going to do is pick up the phone and call 911, you're going to run in the room grabbing the AED on your way in, right, you go in, you're going to check for a pulse and responsiveness, no pulse, no responsiveness, you're going to put the AED pads on and shock this patient immediately and then you're going to start CPR if necessary, so this is an immediately life threatening rhythm, every once in a while this rhythm can occur with a pulse but if this person did have a pulse it would be very threading with the low blood pressure because the ventricles

0:57:56 at a heart rate of 214 beats per minute do not have enough time to fill with any kind of blood, so this is close to three and a half beats per second, there's no way the ventricle has time to fill with blood, so even if they do have a pulse this patient's in a life threatening situation, so this is going to be an important rhythm for you to interpret, you're going to see this on the quiz, you're going to see this on the next hourly exam, you're going to see this on the final

21 Mechanism of Ventricular Fibrillation

0:58:23 – 0:59:57

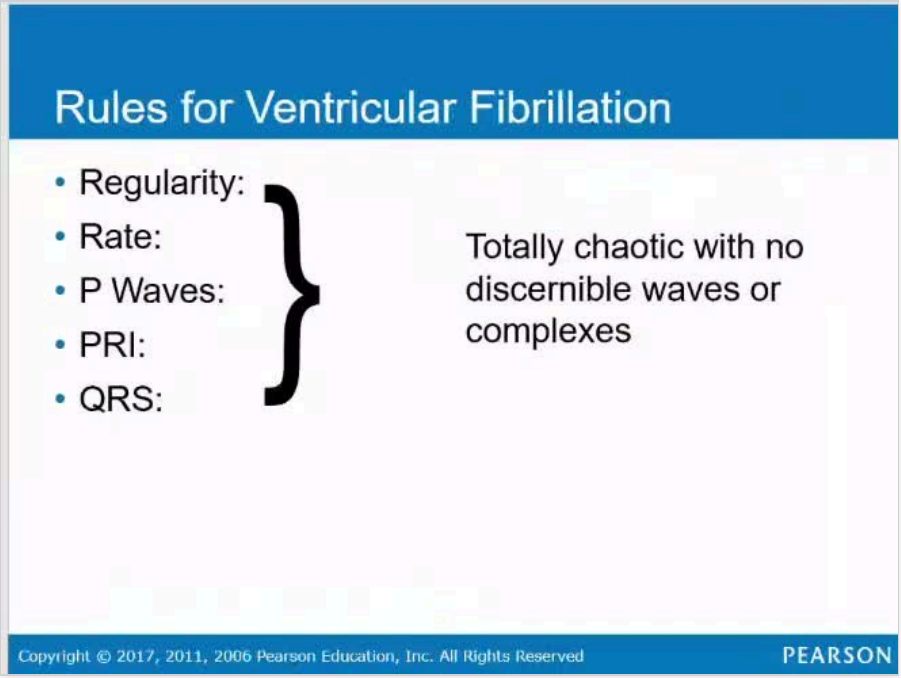


0:58:24 exam. The next rhythm I want to talk about here is ventricular fibrillation and so in ventricular fibrillation now we get irritable focal points but we get multiple irritable focal points as you can see all the green stars, so the suffix i means many, so instead of saying many focal points I'm just going to say multiple foci within the ventricle become irritable and they all generate impulses which makes the heart respond in an uncoordinated chaotic pattern where the heart muscle now just fibrillates instead of contracting,

0:59:14 so there's no way that the heart muscle can respond to that many stimuli at the same time, so it more or less quivers as opposed to contracting, so there is no cardiac output in this rhythm whatsoever, this is immediately life-threatening, again you're going to pick up the phone, call 911, grab the AED, you're going to put the pads on, shock the patient if there's more than one of you that's the other person should start CPR while the other person's putting the pads on, you guys are all going to go through BLS before you graduate and you'll learn how to do all of these things that I'm talking about, so let's take a look at the rules, there's basically

22 Rules for Ventricular Fibrillation

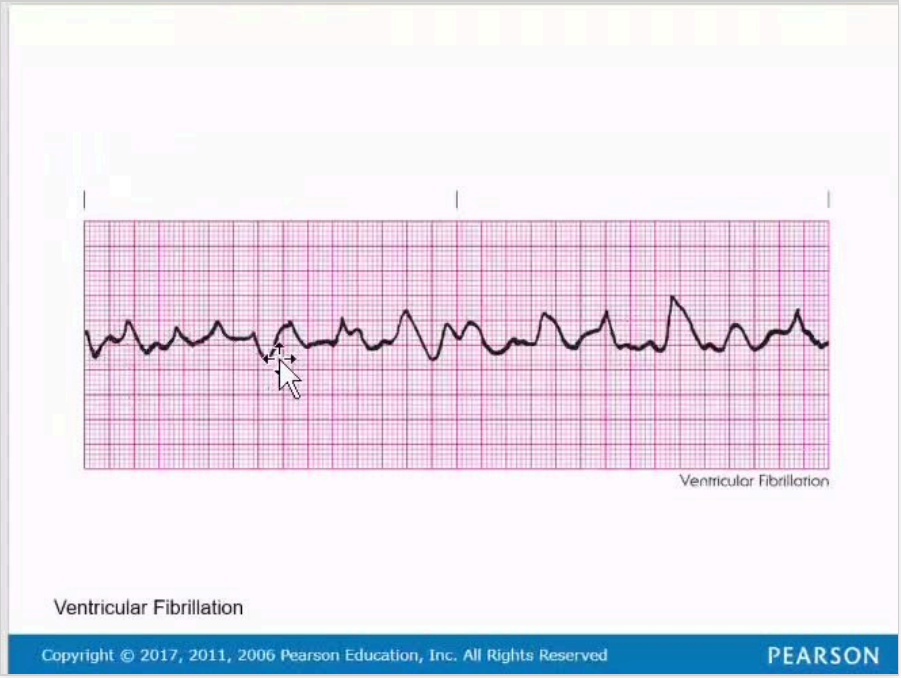
0:59:57 – 1:00:25



1:00:00 a chaotic electrical pattern on this rhythm strip which is completely irregular and that's why we call it chaotic, so the term I would expect you to use would be chaotic for the regularity, there's no heart rate, there's no P waves, there's no PRI, there's no QRS, so let's take a look at what the rhythm looks like, so again you

23 Ventricular Fibrillation (ECG strip)

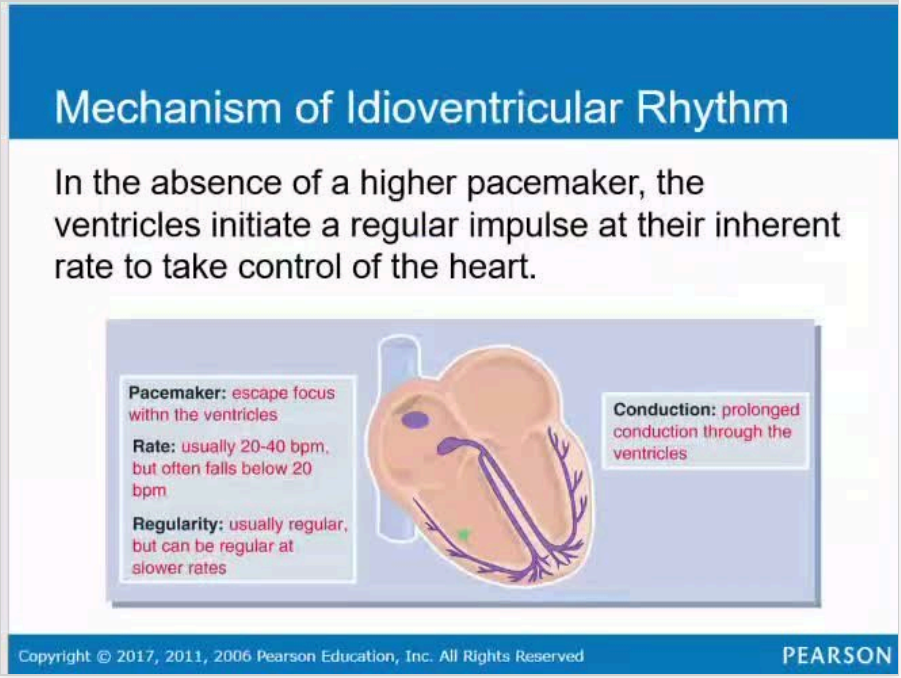
1:00:25 – 1:01:02



1:00:28 can see this chaotic waveform that is completely inconsistent as you go across the strip, so the regularity here is going to be chaotic, the heart rate is none, P waves none, PRI none, QRS none and you have to immediately recognize this as ventricular fibrillation and sometimes we'll call this V-fib short for ventricular fibrillation, again immediately life-threatening. All right we have two more rhythms to cover in this chapter, I know it's a lot, you've

24 Mechanism of Idioventricular Rhythm

1:01:02 – 1:03:02



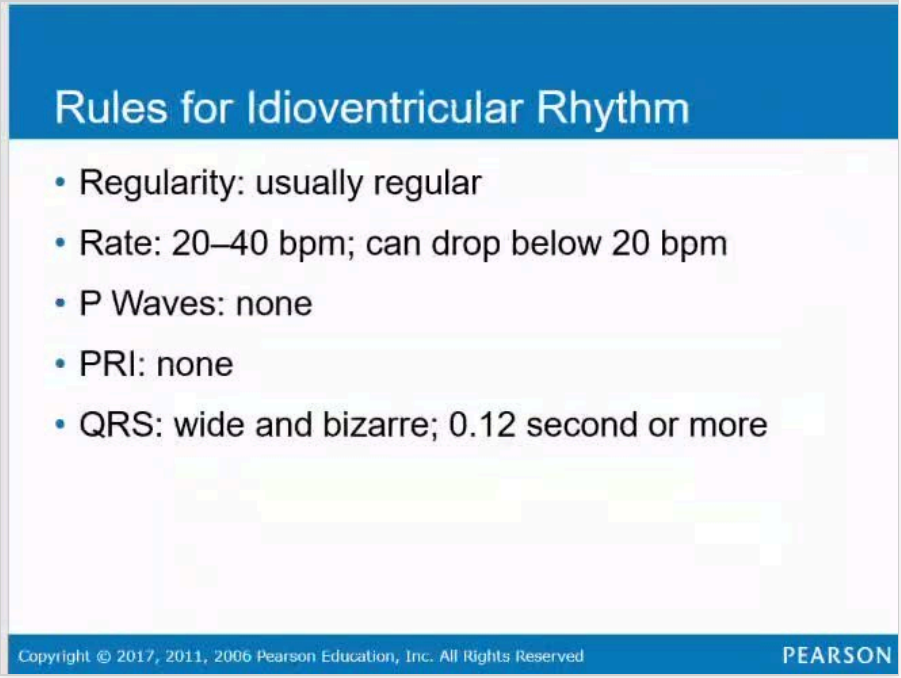
1:01:04 got to stick with it, you're going to have to do some work, put in some time to be able to interpret all of the different things that occur with ventricular rhythms. So our next rhythm is going to be idioventricular, so this is actually an escape mechanism. In the beginning of the semester we talked about the SA node is the primary pacemaker, the AV node is the secondary pacemaker, so something happened to the SA node, the AV node would be next in line to pace the heart. Well in this particular case if something happens to the SA node and the AV node, now we're left to somewhere down in the ventricles as the last backup pacemaker to actually have a heart rate and contract blood. So the mechanism of pacemaker that kicks in in order to keep somebody alive. So again you can see the green star down here, what's happening in this rhythm is the SA node fails, the AV node fails, and so without

1:02:18 any electrical impulses now somewhere down here in the ventricle is going to start kicking in and causing impulses for the heart to depolarize. The problem with this escape ventricular escape mechanism is the inherent rate for the ventricles is 20 to 40, so it's generally not enough cardiac output at a heart rate between 20 to 40 to sustain an adequate blood pressure.

1:02:48 So this rhythm is also life-threatening, it's going to be important that you able to recognize it fairly quickly and then act or respond appropriately. So let's take a look at the rules. So for the regularity or the

25 Rules for Idioventricular Rhythm

1:03:02 – 1:03:42

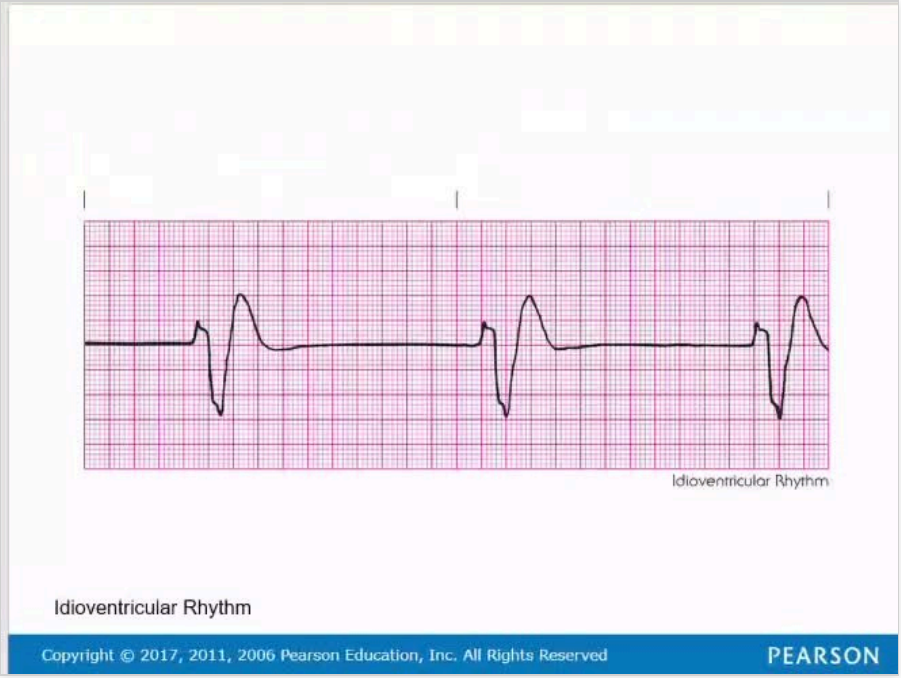


1:03:06 R to R spacing it's usually going to be regular. Generally the heart rate is going to be between 20 to 40, it can be a little slower than that.

1:03:19 If it does, if it's less than 20 we typically call it an agonal rhythm, that means the heart is kind of dying. There will not be any P waves because this is a ventricular rhythm. There'll be no PRIs and the QRSs will be wide and bizarre, R and T going in the opposite direction. So let's take a

26 Idioventricular Rhythm (ECG strip)

1:03:42 – 1:07:23



1:03:42 look at this. This is the last backup pacemaker. So when we look at this, there's no electrical activity until this first beat. This is not a P wave, this is part of the ventricular contraction. So remember that spot in the ventricle is what is initiating these impulses that create ventricular depolarization. Can't be any P waves because the impulse is coming from the ventricles. So this is part of the QRS and then we get this little notch right here and then we have T wave and then we have this long pause till we get to the next beat and then this pattern repeats itself. So let's go through our information gathering. We're going to stay consistent on how we gather information. So this R wave is right on the line, it helps us our counting become a little bit easier. So I'm going to go 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 56, 57 and a half. So there's 57 and a half small boxes between the

1:05:01 first and the second R wave. 5, 10, 50, 20, 25, 30, 35, 40, 45, 50, 55. So this one is 57 and a half and this one is 55. So the spacing is greater than one and a half small boxes. So the regularity here, I'm going to call this slightly irregular because it's greater than one and a half small boxes. So because it's slightly irregular, now I'm going to count my heart rate by the number of R waves multiplied by 10. 1, 2, 3. The heart rate for this rhythm is 30 beats per minute.

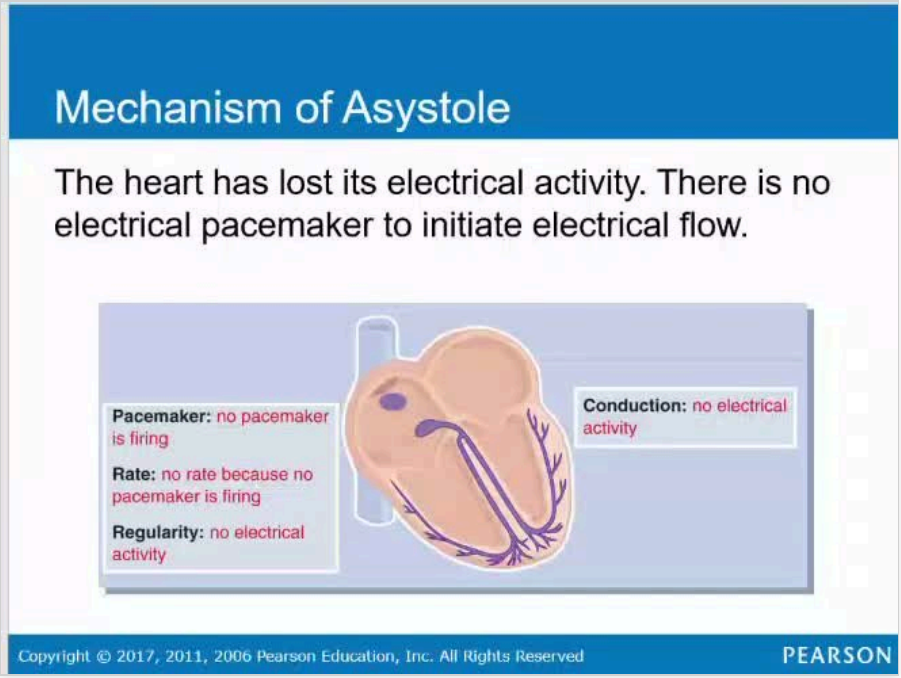
1:05:51 There's no P waves so I can't measure the PRI. For the QRS, we're going to measure from there to that little notch right there. So that ends right here. So again, I'm going to count backwards. That's 1, 2, 3, 4, 5, 6, 7 and a half. So I think this starts right in between. So I'm

going to start with a half and when I get to here, that's 5 and a half, 6, 7 and a half. So this QRS is 7 and a half small boxes. So that's going to get me to 0.30 for my QRS. I believe I'm going to go 7 times 0.04, 7 times 0.04 plus 0.02 gets me to 0.3 seconds.

1:06:57 And then my interpretation here is because I have nothing but wide and bizarre QRS ventricular beats, my interpretation here could be one of two things, idio ventricular rhythm or ventricular escape. All right, one more rhythm in this ventricular chapter. So now

27 Mechanism of Asystole

1:07:23 – 1:08:16



1:07:28 we're going to talk about asystole. So the first thing I want to talk about here is in the name. The prefix A means without. So when we say asystole, it means without systole, without contraction, without depolarization. So there's no electrical activity in this because remember EKGs are electrical activity recorded over time. So if there's no depolarization or no contraction, there's no electrical activity. So this is also known as flat line. So let's look at the rules. There will not be any beats. It will not have a rate or a P wave or a PRI or a QRS.

28 Asystole (ECG strip)

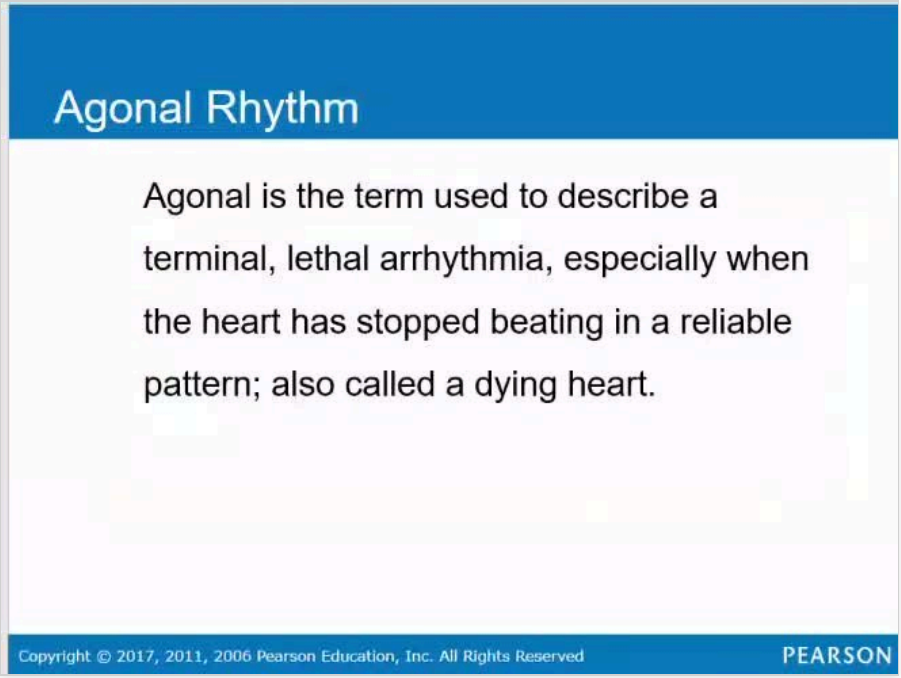
1:08:16 – 1:09:19



1:08:16 And so this is what asystole looks like. So it's basically flat. No electrical activity is what this represents. So there's no pulse, no heartbeats, no cardiac output. This is a life-threatening rhythm. Should be picking up the phone, calling 911, and then you start CPR. Again, you'll learn that this is probably not a, it's not a shockable rhythm. Although if you think it's VFib, you could try to shock, but the AED will say, you know, no shock advice, continue CPR. So I will have, you will have to recognize asystole on quizzes, tests, and final exam as you will ventricular escape rhythm, ventricular fibrillation, ventricular tachycardia, and all the multiple patterns of PVCs that we covered. All right, last slide in the chapter here, we're going to

29 Agonal Rhythm

1:09:19 – 1:10:42



1:09:20 talk about agonal rhythm. And that agonal rhythm is really the idioventricular or ventricular escape rhythm, where the heart rate is less than 20. So agonal is termed, is a term used to describe a terminal lethal arrhythmia when the heart has stopped beating in a reliable pattern, also called a dying heart. So this is ventricular escape rhythm when the heart rate gets below 20.

1:09:55 Okay, so I also just want to put out there that I do have office hours available on Thursdays and Fridays by appointment. You guys are usually on campus on Thursdays, so it's probably easier to come and see me then. I have pharmacology for the respiratory students at one o'clock to three o'clock, but generally I can squeeze in some time before that class. So you know that my office is 208, so when you guys are there on campus, you know, send me an email, say, hey, Mike, do you and I'll help you out as best I can. All right, that concludes our ventricular rhythm lecture.